Can Prompting Reliably Close the Sentence-Length Dispersion Gap?

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Abstract

In machine-text detection, burstiness is the variation of perplexity across a document; in everyday use it is the variation of sentence length, whose obvious statistic, the variance of sentence length, grows with mean sentence length. We give the sentence-length reading a scale-free definition: sentence boundaries as a point process in word-time, scored with the Goh–Barabási burstiness coefficient under the Kim–Jo finite-n correction, bounded and with no model in the loop. The protocol is paired evaluation on 200 era-, topic-, and register-matched Wikinews articles dated 2016–2021: five prompting modes, nine models, three samples per headline and mode, uniform normalization of generated text before scoring. Scored raw, markdown scaffolding manufactures dispersion: on the audited model it produces an equivalence verdict that normalization reverses (instruction-mode 0.006 scored raw, equivalent at the 0.05 margin; 0.035 on normalized text, significantly different, $p = 2 \times 10^{-5}$, and equivalent only from $m^* = 0.052$ upward), and its uninstructed gap grows from 0.053 to 0.102. Uninstructed, eight of the nine models write significantly flatter than the human articles, a median paired gap of 0.050 to 0.128 in B_n. Of the forty-five model-by-mode cells, thirty-two land significantly flatter than the human articles, nine show no significant difference, and four land significantly more dispersed: machine flatness is a strong central tendency that the prompts tested move but do not reliably close. Under the same instruction, five models remain significantly flatter than the human articles, three show no significant difference and one writes significantly more dispersed text; three of the nine are equivalent within ± 0.05 and five within ± 0.08, two of them while differing significantly (none within ± 0.03), and for one model among the nine the instruction makes the text significantly flatter than its own uninstructed output. Few-shot exemplars of typical human news prose produce text significantly flatter than the instruction alone in every model.

1 Introduction

Burstiness is one of the signals most often cited to separate human writing from machine writing [Fraser et al., 2024], and the word carries two readings. In detector practice it means the variation of perplexity across a document, part of the sense GPTZero gave it [Tian, 2023]. In everyday use it means the variation of sentence length, the mix of long and short sentences that gives prose its rhythm and that language models flatten toward a middle. The obvious statistic for that reading, the variance or standard deviation of sentence length, grows with mean sentence length, so uniformly long sentences score burstier than varied short ones. This paper is about the second reading.

Sentence-length dispersion has been a stylometric variable since Yule [1939] and Williams [1940], but its natural statistics, the variance and the standard deviation of sentence length, grow with the mean, and the scale-free ratio Tarım and Onan [2025] use is not corrected for the few sentences a

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document has. The aim of this paper is to give the sentence-length reading a statistic that is both scale-free and corrected, and to use it to benchmark five prompting techniques.

A definition. We treat sentence boundaries as a point process in word-time and score the sentence-length sequence with the Goh–Barabási burstiness coefficient [Goh and Barabási, 2008] under the Kim–Jo finite-n correction [Kim and Jo, 2016], written B_n. It is bounded in $[-1, 1]$, needs no reference model, and is a function of the coefficient of variation and the sentence count alone, so it is scale-free: scaling every sentence length leaves it unchanged, and dispersion is measured relative to typical sentence length instead of conflated with it. Alongside B_n we report a Gini robustness check [Gini, 1912], Miller–Madow-corrected punctuation entropy as a separate segmentation axis, and a memory coefficient M for lag-1 order; surprisal, the strong but model-relative detector signal, stays deliberately outside the construct. Because a metric that becomes a target invites gaming, we add a Goodhart guard, a screen that flags degenerate constructions that inflate any dispersion score, for any use that selects text by the score; a quality gate in the sense of Shypula et al. [2025] is the second condition such a use would need.

A benchmark of five prompting techniques. We evaluate control (a bare brief), instruction, few-shot human exemplars [Brown et al., 2020, Min et al., 2022], an outline-then-write plan mode [Wang et al., 2023], and a draft-critique-rewrite self-refine loop [Madaan et al., 2023] against 200 English Wikinews articles [Wikinews, 2021] dated 2016–2021, with era, topic, and register held fixed. Each generation is length-matched to its paired article, and every cell is sampled three times. Generated text is uniformly normalized before scoring, because markdown scaffolding and echoed headlines in raw output manufacture dispersion the human baseline lacks, and on the audited leg this choice moves the uninstructed dispersion gap from 0.053 to 0.102. The identical design runs across nine models, spanning open-weight checkpoints (Qwen3.6-27B, Gemma 3 27B), API-served models (DeepSeek V4 Pro, GLM-5.2, Kimi K2.6, Grok 4.3), and closed frontier systems (Grok 4.5, Gemini 3.1 Pro, Claude Opus 4.8).

The aim problem. Prompting changes sentence-length dispersion, and the landing depends on the model. Uninstructed, eight of the nine models write significantly flatter than the human articles, with control deficits 0.050 to 0.128; the three largest belong to Gemini 3.1 Pro, GLM-5.2 and Claude Opus 4.8. Under the identical instruction, five remain significantly flatter, three show no significant difference and one writes significantly more dispersed text; three of the nine are equivalent within ± 0.05 and five within ± 0.08, two of them while differing significantly (none within ± 0.03), and for one model among the nine the instruction makes the text significantly flatter than its own uninstructed output; a model's uninstructed gap does not predict where it lands. Few-shot exemplars of typical human news prose produce text significantly flatter than the instruction alone in every model.

Contributions.

1. A reproducible definition of sentence-length burstiness that is model-free and scale-free: the Goh–Barabási coefficient under the Kim–Jo finite-n correction, with companions (Gini check, segmentation-diversity axis, memory coefficient; surprisal kept outside) and a Goodhart guard proposed for any use that selects text by the score (Section 3).

2. A paired protocol on era-, topic-, and register-matched news articles: uniform normalization before scoring, seed-averaging that avoids pseudo-replication, within-model multiplicity correction, equivalence only as a CI inside pre-specified margins, a contamination probe, and a moderation-refusal audit (Sections 4.1–4.4 and 6).

3. The aim problem across nine models, five modes, three samples per cell: every mode-to-mode claim the text builds on carries a direct paired contrast with rankings marked descriptive, and few-shot exemplars produce text significantly flatter than the instruction alone in every model (Section 4).

2 Related work

2.1 Burstiness in detection

In detector practice the term comes from GPTZero, whose original detector paired perplexity with a burstiness scalar, the variation of writing patterns and perplexity across the document [Tian, 2023], and which recent attribution work uses as a baseline detector [Radvand et al., 2025]; the survey that cites it describes burstiness as variability in style, tone and vocabulary [Fraser et al., 2024]. The sentence-length reading travels under the same name: Tarım and Onan [2025] define burstiness as the variance of sentence-level perplexities and compute it as the coefficient of variation of sentence lengths, a ratio that is scale-free but uncorrected for the number of sentences; Section 3 gives the corrected form. DivEye [Basani and Chen, 2025] tracks how unpredictability fluctuates across a text, the nearest text-side analog to timing variation. Surveys [Fraser et al., 2024] treat perplexity-burstiness as a standard pair and document detector vulnerability to paraphrasing.

2.2 Stylistic variation and diversity

The stylistic-variation benchmark [Milička et al., 2025] compares frontier models' counterparts of human reference texts with those texts using Biber's multidimensional analysis, a register-level comparison rather than one of sentence-length dispersion. Effective semantic diversity [Shypula et al., 2025] gates diversity on quality; a gate of that kind is the condition we would attach to any use of B_n that selects text, so that incoherent high-variance text is not rewarded.

3 Formalizing burstiness

We give the sentence-length reading of burstiness (Section 2) a model-free, reproducible instrument.

3.1 Burstiness as a point process

Let a document D be segmented into sentences with word-length sequence $L = (l_1, \ldots, l_n)$. We treat the sentence boundaries as a point process in word-time, so L is a sequence of inter-event intervals, and score it with the Goh–Barabási burstiness coefficient [Goh and Barabási, 2008],

$$B = \frac{\sigma - \mu}{\sigma + \mu},$$

where μ and σ are the mean and standard deviation of L. B is bounded in $[-1, 1]$: $B = -1$ for a metronomic sequence, $B = 0$ for a Poisson one, and $B \to 1$ for a maximally bursty one. Because real documents give short interval sequences, we apply the Kim–Jo finite-n correction [Kim and Jo, 2016]: with $r = \sigma/\mu$ and n sentences,

$$B_n = \frac{\sqrt{n+1}\, r - \sqrt{n-1}}{(\sqrt{n+1} - 2)\, r + \sqrt{n-1}},$$

which reduces to $B = (r-1)/(r+1)$ as n grows and removes the small-n bias of B.

B_n is *scale-free*: it depends on the sentence lengths only through r and n, so scaling every length by the same factor leaves it unchanged, and it measures variation *relative* to typical sentence length rather than conflating dispersion with length. Classically, B is a bounded reparametrization of the coefficient of variation, weak alone in the stylometric literature [Yule, 1939, Williams, 1940]. What is new is the corrected, model-free formulation, its paired evaluation on a matched modern corpus, and the nine-model prompt benchmark built on it (Section 4).

3.2 The construct and its companions

Burstiness proper is the single scale-free coefficient B_n. We report four companions alongside it, none folded into the construct:

- **Gini robustness check.** The Gini coefficient [Gini, 1912] of the sentence lengths is a scale-free dispersion measure that aggregates over all pairwise differences, so it is less hostage

to a single long sentence. It agrees with B_n in direction on every cell of the nine-model benchmark where either difference is significant (Section 4); agreement shows a result is not an estimator artifact.

- **Segmentation diversity, a separate axis.** The Miller–Madow-corrected Shannon entropy of the punctuation distribution measures how varied the segmentation symbols are, a distinct axis from length dispersion. Typographic quotes are mapped to ASCII before the punctuation count and before sentence splitting (Appendix D), so smart-quote typography registers neither as diversity nor as a missed boundary.
- **Memory, the order term.** The Goh–Barabási memory coefficient M [Goh and Barabási, 2008], the lag-1 correlation between consecutive interval lengths plus $1/(n-1)$, which offsets the lag-1 estimator's small-sample bias of about $-1/(n-1)$ (Appendix C), is the order information a dispersion statistic discards. In Section 4 six models' few-shot legs overshoot the human articles on memory while undershooting on dispersion.
- **Predictability, kept outside.** Mean per-token surprisal and the mean absolute consecutive difference, under a fixed reference model, form the strong but model-relative detector signal. The definition places them *outside* the construct: they are a first-moment quantity, whereas burstiness is a second moment of a derived sequence. The benchmark scores the four model-free dimensions only and reports no surprisal values.

We compute raw sentence-length variance $\mathrm{var}(L)$ only as a diagnostic; it is confounded with absolute length by construction, the central defect of the colloquial statistic. Appendix D scores the benchmark with it beside B_n (Table 14), and its verdicts follow length as well as dispersion.

3.3 Anti-gaming screens and reproducibility

A metric that becomes a selection score invites gaming. Degenerate texts can reach a high-dispersion score (one giant sentence among fragments, mechanical alternation, punctuation spam). A Goodhart guard flags these; the full flag list ships in the reference implementation (Appendix C). The implementation marks documents below ten sentences as unreliable; the benchmark scorer requires five, and Appendix D reports how many scored generations fall between (at most 23% of a leg's generations in any one mode) and that a ten-sentence floor changes one of the 45 landings re-scored. Re-punctuating a fixed text can move B_n across much of its bounded range, so the structural metric must never select generations alone; any use that selects by it needs a quality gate in the sense of Shypula et al. [2025], which we have not built. The benchmark of Section 4 is descriptive and selects nothing by the metric, so the guard does not filter the scored generations; it fires on 4% of the human articles and 2–14% of a leg's generations in a given mode, and re-scoring without the flagged text changes one of the 45 landings re-scored (Appendix D).

The metric, the guard, and the multiscale profile the guard's wash-out check reads are pure standard library in `experiments/burstiness_pp.py`, with a self-check; Appendix C reproduces the full implementation. They run without a model; of the companions above, only surprisal would need one, one more reason it stays outside the construct.

4 The aim problem: how reliably instructions move the construct

4.1 Corpus and generation

The corpus is 200 English Wikinews articles [Wikinews, 2021] (CC BY 2.5; released with the model generations [Lanzo and Angelo, 2026]), dated 2016–2021 and therefore pre-model. The model receives the headline and the paired article's word count, so topic, register and target length are held fixed within a pair and the human side is pre-model; the facts and quotations are the model's own (achieved lengths in Appendix D). The human side of each pair is the plain article body, regenerable from the source Wikinews wikitext by a deterministic parse (`experiments/ablation/clean_wikinews.py`) that, on a held-out check of 100 articles, reproduces the shipped human bodies exactly. The first leg is a frozen open-weights Qwen3.6-27B (bf16, temperature 0.7, top-p 0.95, reasoning mode enabled, reasoning stripped before scoring). Eight further models join in Section 4.4. For each headline the model writes a news article from a brief containing only the headline and the paired human article's word count, so each generation targets its pair's length. We compare

five non-overlapping prompting modes: a *control* brief with no stylistic instruction, an *instruction* mode that adds a metric-free persona directive to write as a professional reporter in a natural human voice, never flat or mechanical, restated near the end of the prompt, *few-shot* exemplars of matched human news prose [Brown et al., 2020, Min et al., 2022], an outline-then-write *plan* mode [Wang et al., 2023], and a draft-critique-rewrite *self-refine* loop [Madaan et al., 2023]. The three few-shot exemplars are representative human news reports, matched by register and topic and centered on the median B_n of the exemplar pool (each near $B_n = -0.416$; the evaluation set's median is -0.414). The selection uses B_n only to find the exemplars nearest that median, never the high-dispersion end, so the few-shot mode sees typical human sentence-length dispersion. No prompt names the statistic or gives a numeric target on it. Every non-control prompt asks for prose that is never flat or mechanical; the few-shot and self-refine prompts name rhythm, pacing, flow and cadence, and the rewrite step says to change how the sentences move, as a user asking for human-sounding prose would. We sample each (mode, headline) cell three times, at fixed seeds on the open-weight legs and as independent calls on the API legs, to which no seed is sent; we call a cell's three samples its seeds throughout. Of the leg's $3{,}000$ cells, $2{,}996$ completed; we excluded the four that exceeded the reasoning budget rather than scoring their incomplete output. We average affected headlines over their remaining seeds, so every mode retains 200 matched pairs. The entire run is scripted, and every reported number can be recomputed from the released artifacts.[1] Appendix A lists the verbatim wording of every mode and the rule that composes them.

4.2 Normalization: stripping the wrapper before scoring

Model output does not arrive as plain prose. A generation-hygiene guard flagged markdown scaffolding and echoed headline blocks in every model we benchmark, from 18% (markdown) and 40% (title lines) of control-mode generations on one leg (DeepSeek V4 Pro) to 100% on another (Gemini 3.1 Pro, both features). All of it is formatting the plain-prose human baseline lacks. A segmenter fed raw output scores a markdown header or an echoed dateline as a short pseudo-sentence, and those fragments manufacture length dispersion and segmentation symbols the article prose does not have. Every generated leg is therefore normalized before scoring: a deterministic script (`experiments/ablation/normalize_generation.py`) drops the echoed headline block and unwraps markdown syntax while preserving the article prose; the human baseline is untouched. A word-loss audit of the full first leg finds median 0.0%, mean 1.7%, and at most 7.4% of a generation lost to normalization, that maximum being the headline block itself.

Scaffolding sits outside the construct: B_n is defined over the sentence lengths of prose, and a markdown header or an echoed dateline is not a sentence of the article. Scoring it credits formatting as dispersion, the same class of inflation the Goodhart guard flags (Section 3), so normalization is a corpus-level Goodhart correction, applied uniformly to every leg. Its size is measurable on the audited leg (Qwen3.6-27B): scored raw, instruction-mode B_n is equivalent at the 0.05 margin (0.006, CI $[-0.014, 0.031]$, $m^* = 0.031$, $p = 0.10$), where normalized text differs significantly (0.035, CI $[0.010, 0.052]$, $p = 2 \times 10^{-5}$) and its interval needs a margin of 0.052 to count as equivalent; a raw control-mode gap on memory (-0.063, $p = 0.003$) is not significant on normalized text (-0.017, $p = 0.20$), and segmentation entropy shows no significant control-mode difference under either scoring, which is why we describe that leg's control gap as significant only on the dispersion dimensions; and the control-mode dispersion gap on normalized text is 0.102 against a raw estimate of 0.053, with per-seed rank-biserial rising from 0.27–0.38 to 0.61–0.68.

4.3 Analysis

Because the three seeds of a cell share their human counterpart, pooling them as separate pairs would pseudo-replicate the human documents and overstate significance. We therefore average each headline's metric vector across its seeds first and run one Wilcoxon signed-rank test per mode and dimension over the matched pairs (197–200 per model and mode). Matched rank-biserial is reported as effect size. Each test compares one mode against the human articles; where the text compares

[1] Scoring code, prompts, the metric implementation, per-model scored evaluations (normalized protocol, plus the audited leg's raw-protocol archive) and generation metadata accompany the paper; the full repository, with the research database, is at `https://github.com/VittoriaLanzo/burstiness-was-measured-wrong-lp4fm`, and the human articles and the raw generations of every leg and of the paraphrase probe are released as one dataset [Lanzo and Angelo, 2026].

two modes of the same model, it cites a direct paired contrast between them on the same headlines (Appendix D; contrast deltas are in the model's own B_n; negative means the first mode is flatter).

Every model ran the identical twenty-cell grid (five modes by four dimensions), corrected within model with Benjamini–Hochberg at $q = 0.05$, Holm reported alongside; the correction script and each leg's census ship with the released artifacts. Across the nine models' grids, 133 of the 180 cells are significant at the per-cell 0.05 level; 132 survive Benjamini–Hochberg and 116 survive Holm. The one cell that falls under Benjamini–Hochberg (Kimi K2.6's control-mode segmentation entropy) is not on B_n, and the prose does not build on it. No correction is applied across models; Table 1 carries per-cell stars with a pointer to each model's corrected census in Appendix B.

Matching claims are never stated as failures to reject. A match requires the 95% CI of the median paired delta inside a pre-specified margin; margins 0.03, 0.05, and 0.08 were fixed before the cross-model runs. We also report m^*, the smallest margin the CI would clear, the larger of the two absolute CI bounds. CIs on the median paired delta are percentile bootstrap intervals: the 2.5th and 97.5th percentiles of the medians of $2{,}000$ resamples of the matched pairs. Significance and equivalence are separate verdicts: a delta can differ significantly from zero and still sit inside a margin, and the text reports both. We say a prompt closes a model's gap when its cell is both not significantly different from the human articles and equivalent at a pre-specified margin: equivalence alone would count a significant gap as closed. For scale, the 200 human articles have median $B_n = -0.414$ with standard deviation 0.127, so the three margins are 0.24, 0.39 and 0.63 of a human standard deviation.

4.4 One instruction for nine models

The design of Section 4.1 runs identically on every model: the same 200 headlines, the same five wordings, three samples per cell, uniform normalization, the same paired scoring. The nine models are Qwen3.6-27B and Gemma 3 27B (open weights, bf16, temperature 0.7, top-p 0.95); DeepSeek V4 Pro, Grok 4.3, GLM-5.2, and Kimi K2.6 (API-served, provider-default decoding; GLM-5.2 ignores a requested temperature and Kimi K2.6 rejects one); and Grok 4.5, Gemini 3.1 Pro, and Claude Opus 4.8 (closed APIs; Grok 4.5 exposes neither temperature nor top-p, Gemini 3.1 Pro ran at temperature 1.0 with provider-managed thinking, and Claude Opus 4.8 exposes no temperature). Thinking modes, effort levels and output caps are recorded in each leg's released metadata and listed in Appendix D. Two of the nine models' API legs used a $32{,}000$-token cap instead of $16{,}000$: DeepSeek V4 Pro from early in its run (Appendix D.3), after reasoning exhausted the smaller cap, and Claude Opus 4.8, because Anthropic bills reasoning against the cap, so at maximum effort the self-refine rewrite spent the whole budget reasoning and emitted no prose. That leg ran at effort high (Gemini 3.1 Pro ran at thinking level high) and completed all $3{,}000$ cells with no truncated or empty output. A tenth leg, labeled DeepSeek V4 Pro† throughout, re-ran the DeepSeek V4 Pro grid on 18 October 2026 with the same prompts and briefs against the snapshot the API alias then served; it is reported beside the nine models in every cross-model table and in Figure 1, has its own grid (Table 9), and stays outside the nine-model counts (Appendix D). Decoding is not matched across providers, so the benchmark reads as nine separate paired comparisons against the same human baseline, each model compared against the human articles alone. Where a provider returns reasoning content separately we strip it before scoring. Gemini's thoughts are never returned, and Gemma 3 has no thinking mode. We benchmark single generators only; multi-agent systems are excluded because they break the one-generator pairing.

Completeness is disclosed per leg. Qwen3.6-27B excludes four cells for exceeded reasoning budgets (Section 4.1), and Grok 4.5 reports two empty self-refine outputs, truncations by signature, seed-averaged away. Provider content moderation removes cells on two legs, by two different mechanisms. Moonshot refuses explicitly: 25 of Kimi K2.6's $3{,}000$ cells return HTTP 400, one Hong Kong protest headline refused at every mode and seed. Zhipu refuses silently: 36 of GLM-5.2's cells return an ordinary HTTP 200 with empty output, concentrated on the same cluster of politically sensitive headlines. The harness records both as empty cells and seed-averaging excludes them, so every paired statistic is computed over the headlines the model actually wrote (197–200 per mode for GLM-5.2, 198–199 for Kimi K2.6; Appendix B); a benchmark that does not audit empty outputs would silently absorb the second mechanism into its scores. The remaining five legs, and DeepSeek V4 Pro†, generated all 200 headlines in every mode.

Table 1: The cross-model aim problem on dispersion: median ΔB_n = human − model per mode, seed-averaged Wilcoxon paired tests over the matched headlines each mode completed (197–200; Appendix B). Positive places the model below the human articles (flatter), negative above them (more dispersed). Matched rank-biserial in parentheses; $^{*}p < 0.05$, $^{**}p < 0.01$ per cell; each model's corrected twenty-cell grid is in Appendix B, with no correction across models (Section 4.3). The two rows below the rule are successive Grok generations. The row DeepSeek V4 Pro† is the DeepSeek leg re-run on 18 October 2026 against the snapshot the API alias then served, with the same prompts and briefs (Appendix D); it is reported beside the nine models and outside their counts.

Model	Control	Instruction	Few-shot	Plan	Self-refine
Gemini 3.1 Pro	0.128 (0.86)**	0.071 (0.56)**	0.169 (0.94)**	0.071 (0.58)**	0.039 (0.28)**
GLM-5.2	0.126 (0.85)**	0.064 (0.58)**	0.148 (0.94)**	0.081 (0.66)**	0.012 (0.15)
Claude Opus 4.8	0.119 (0.80)**	−0.037 (−0.25)**	0.039 (0.34)**	0.000 (0.01)	−0.065 (−0.50)**
Qwen3.6-27B	0.102 (0.70)**	0.035 (0.35)**	0.119 (0.80)**	0.043 (0.42)**	0.041 (0.27)**
Kimi K2.6	0.106 (0.73)**	−0.007 (0.00)	0.133 (0.85)**	0.029 (0.25)**	−0.069 (−0.52)**
Gemma 3 27B	0.091 (0.76)**	0.057 (0.47)**	0.081 (0.68)**	0.070 (0.62)**	0.045 (0.41)**
DeepSeek V4 Pro	0.050 (0.48)**	0.003 (0.06)	0.032 (0.36)**	0.012 (0.16)	−0.002 (0.04)
DeepSeek V4 Pro†	0.039 (0.34)**	−0.037 (−0.27)**	0.018 (0.14)	−0.030 (−0.18)*	−0.042 (−0.29)**
Grok 4.3	0.052 (0.45)**	0.065 (0.57)**	0.116 (0.83)**	0.082 (0.65)**	0.039 (0.35)**
Grok 4.5	0.002 (−0.01)	−0.022 (−0.13)	0.106 (0.77)**	0.004 (0.05)	−0.034 (−0.30)**

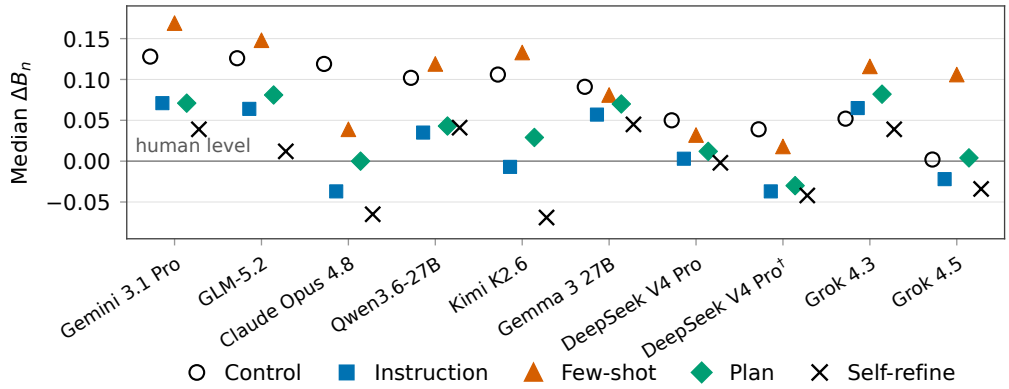


Figure 1: Median ΔB_n = human − model for every model and prompting mode, the forty-five nine-model cells of Table 1 and the five cells of its DeepSeek V4 Pro† row. Reading key: the gray line is the human dispersion level; markers above it are flatter than the human articles, markers below it more dispersed. Few-shot sits highest in six of nine models by median and is the largest-effect mode by rank-biserial in six, and instruction lands on both sides of the line, including across the three leftmost models whose control markers nearly coincide. Model order and significance: Table 1.

Table 1 reports the B_n cell for every mode on every model, Figure 1 plots all forty-five cells and the five cells of DeepSeek V4 Pro†, and Appendix B carries each model's full four-dimension grid with its correction census.

Uninstructed, eight of nine models write significantly flatter than the human articles, with control deficits from 0.050 to 0.128; Grok 4.5's control-mode difference is not significant (0.002, $p = 0.86$). The three largest gaps belong to Gemini 3.1 Pro (0.128, rank-biserial 0.86), GLM-5.2 (0.126, 0.85) and Claude Opus 4.8 (0.119, 0.80). Deficits of 0.050 to 0.128 are 0.4 to 1.0 human standard deviations: the model distributions overlap the human one rather than sitting outside it.

Under instruction, four models write significantly more dispersed text than under their own bare brief (paired contrasts in Appendix D) while remaining significantly flatter than the human articles: Gemini 3.1 Pro lands at 0.071, GLM-5.2 at 0.064, Qwen3.6-27B at 0.035 and Gemma 3 27B at 0.057.

Table 12 gives both verdicts for every model under instruction: whether the median delta differs significantly from zero, and the tightest pre-specified margin its 95% CI clears. Three models show

no significant difference. Equivalence, cumulative across margins, holds at 0.03 for none of the nine (DeepSeek V4 Pro's upper bound, 0.0302, sits on that margin; Appendix D), at 0.05 for three and at 0.08 for five, and a model can differ significantly and still be equivalent at a margin. DeepSeek V4 Pro closes a significant control gap (0.050, rank-biserial 0.48, $p < 10^{-5}$) to 0.003 (contrast 0.057, $p < 10^{-5}$) with CI $[-0.023, 0.031]$ ($m^* = 0.031$) on the July snapshot, whereas the 18 October snapshot, a different checkpoint behind the same alias (DeepSeek V4 Pro†), lands significantly past the human articles under the same wording (-0.037, $p = 8 \times 10^{-4}$, equivalent at the 0.08 margin; Appendix D); Kimi K2.6 closes 0.106 (0.73) to -0.007 (contrast 0.101, $p < 10^{-5}$) with CI $[-0.033, 0.017]$ ($m^* = 0.033$). The July DeepSeek landing is on dispersion only: its instructed leg overshoots the human articles on segmentation entropy (-0.145, rank-biserial -0.47, $p < 10^{-5}$) and on memory (-0.057, -0.25, $p = 0.002$) while matching dispersion.

Four of the forty-five cells land significantly past the human articles: Claude Opus 4.8 crosses under instruction (-0.037, rank-biserial -0.25, $p = 0.002$), and self-refine pushes three models past (Kimi K2.6 -0.069, Claude Opus 4.8 -0.065, Grok 4.5 -0.034); all four survive both corrections. Under instruction Grok 4.3, the fifth model still significantly flatter than the human articles (0.065), writes significantly flatter than under its own bare brief (paired contrast -0.019, CI $[-0.028, -0.008]$, $p = 0.001$), while for Grok 4.5 the same instruction moves dispersion the other way (0.015, $p = 0.02$) and leaves it not significantly different from the human articles (-0.022, $p = 0.10$), the one instruction landing that the original segmenter and the guard filter each make significant (Appendix D).

Nine of nine models undershoot significantly under few-shot, and few-shot output is significantly flatter than instruction output in every model (paired contrasts, Appendix D). Against the bare brief the picture is mixed: few-shot is significantly flatter than control in six of the nine models, significantly more dispersed in three and not significantly different in none; ranked by rank-biserial it is the largest-effect mode in six of nine, a descriptive ranking. The exemplars are three human news reports at the median of the human B_n distribution (Appendix A), so showing the model typical human sentence-length dispersion produces text significantly flatter than the instruction alone in every model, and flatter than saying nothing in six of the nine; one reading we did not test is register reproduction rather than variance.

The synthesis is the scatter. Across the forty-five model-by-mode cells, thirty-two land significantly flatter than the human articles, nine show no significant difference, and four land significantly more dispersed. For the prompts tested, prompt-level control has poor aim: the gain and even the sign of an instruction are model-specific, and the uninstructed gap does not predict the landing, so the instruction that reaches equivalence on one model moves the next a different amount and sometimes the wrong way.

5 Discussion

Recommendations for evaluation practice.

1. Normalize before scoring: any segmenter-based feature computed on raw model output measures formatting as well as sentence-length dispersion, and the unreported choice can flip verdicts in both directions (Section 4.2).
2. Sample multiple seeds and average within headline before testing: pooling seeds pseudo-replicates the human baseline and overstates significance (Section 4.3).
3. Audit empty cells: silent provider moderation returns ordinary responses with empty output, and unlogged empties reshape scores invisibly (Section 4.4).
4. State parity as CI-bounded equivalence against pre-specified margins and report the smallest margin cleared; a failure to reject is not a match (Section 4.3).

Sentence-length dispersion as a detection signal. Sentence-length dispersion is the everyday reading of burstiness, and the benchmark says where its flatness signature holds and where it inverts. For a dispersion-based detector, the instructed flatness signature persists in five of nine models, but Claude Opus 4.8, Kimi K2.6 and Grok 4.5 each write significantly more dispersed text than the human articles under at least one prompted mode, where a flatness threshold would miss them, and any segmenter-based feature computed on raw output inherits the formatting artifacts of Section 4.2. The strong detector signal, surprisal, stays outside the construct.

6 Limitations

Limits of the metric. Statistical separation is not construct validity: significant paired differences show the coefficient separates authorship in matched news prose, and they leave open whether it captures everything meant by rhythm. The coefficient B_n inherits the weaknesses of the coefficient of variation it is built on (Section 3): it is order-blind and shape-blind, scoring the dispersion of the length sequence while ignoring where the long and short sentences fall, and memory recovers only lag-1 order. Surprisal is model-relative, which is why it stays outside the construct and is not scored. The reference implementation flags documents under ten sentences; the benchmark scores from five, and Appendix D reports the sensitivity. The metric is genre-relative: target bands do not transfer unchanged across registers. Guard thresholds are heuristic: they set a floor and prove nothing about strategies outside their list.

Limits of the experiment. The corpus is a single register (Wikinews, expository news, English), chosen as a conservative, low-dispersion register, so reported gaps are likely smaller than on narrative or literary text. The nine models occupy a narrow band of contemporary instruction-tuned assistants, with the Grok pair contributing two successive generations; for budget reasons we omitted an OpenAI leg. The flatness tell is model- and snapshot-relative: Grok 4.3 to Grok 4.5 changes the instruction-mode landing in a single generation, and the DeepSeek V4 Pro alias re-queried three months later writes significantly more dispersed text than it did in July in all five modes (Appendix D), so the gaps we measure should not be read as constants of nature. One fixed wording per mode risks measuring one model's reading of one prompt, and neither an explicit instruction to vary sentence length nor a per-model calibration was tested, so the results show how the prompts tested move the gap, not that no prompt can close it. Replication across nine models removes part of the wording risk, since few-shot output is significantly flatter than instruction output in every model. A paraphrase probe samples the other part on one model, the 18 October DeepSeek snapshot, where the canonical wording lands 0.037 past the human level ($p = 8 \times 10^{-4}$): three paraphrases of the same persona frame (one sample per brief, 200 briefs each) land that snapshot 0.047 to 0.067 past, three of three significantly, so no wording we tried returns it to the human level and none, the canonical one included, is equivalent at 0.03 or 0.05. Paired on the same headlines, the paraphrases sit 0.016 to 0.025 beyond the canonical wording (three of three significant). The July snapshot's 0.05 equivalence could not be re-probed: by October the alias served a later snapshot (Appendix D).

Decoding is not matched across providers (of the nine models' legs, four do not honor a temperature setting, two ran at provider defaults, one closed leg ran at temperature 1.0), so cross-model magnitude comparisons carry a decoding confound while within-model comparisons do not, since every mode of a leg ran under the same decoding.

The human articles may appear in the generators' training data: the 2016–2021 corpus predates all nine training cutoffs; we measured overlap directly. An n-gram probe over 600 control-mode generations per model on Qwen3.6-27B, DeepSeek V4 Pro and Grok 4.3 finds mean matched 8-gram overlap with the paired article below 0.003 and 13-gram overlap below 0.001, against a mismatched-article baseline of 0.0 (per-model values in Appendix D). Verbatim reproduction therefore cannot explain the measured gaps on the probed models. Memorization would move dispersion toward the human level, so any unprobed effect, whether a deficit or an overshoot, is, if contaminated, an underestimate. The direction reverses for the equivalence landings: memorization would inflate instructed dispersion to the human level, so contamination could manufacture the equivalences we report. All of them fall in the instruction mode the probe did not cover, and the probe did not cover Kimi K2.6 or Grok 4.5 at all.

7 Conclusion

Burstiness is measured everywhere under two readings, and the sentence-length one has lacked a statistic that is both scale-free and corrected for the number of sentences. This paper gives it one, reproducible and model-free. Its first use is the aim problem: on uniformly normalized text, eight of nine models write significantly flatter than the human articles uninstructed. The same instruction leaves five significantly flatter, three not significantly different and one significantly more dispersed, and makes one model significantly flatter than its own bare brief. The gap is one that the prompts tested move but do not reliably close.

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References

Advik Raj Basani and Pin-Yu Chen. Diversity Boosts AI-Generated Text Detection. arXiv, 2025. URL https://arxiv.org/abs/2509.18880.

Tom B. Brown, Benjamin Mann, Nick Ryder, et al. Language Models are Few-Shot Learners. NeurIPS 2020, 2020. URL https://arxiv.org/abs/2005.14165.

Kathleen C. Fraser, Hillary Dawkins, and Svetlana Kiritchenko. Detecting AI-Generated Text: Factors Influencing Detectability with Current Methods. arXiv, 2024. URL https://arxiv.org/abs/2406.15583.

Corrado Gini. *Variabilità e Mutabilità*. C. Cuppini, Bologna, 1912. URL https://ui.adsabs.harvard.edu/abs/1912vamu.book.....G/abstract.

Kwang-Il Goh and Albert-László Barabási. Burstiness and memory in complex systems. *EPL (Europhysics Letters)*, 2008. doi: 10.1209/0295-5075/81/48002. URL https://doi.org/10.1209/0295-5075/81/48002.

Eun-Kyeong Kim and Hang-Hyun Jo. Measuring burstiness for finite event sequences. *Physical Review E*, 2016. doi: 10.1103/PhysRevE.94.032311. URL https://doi.org/10.1103/PhysRevE.94.032311.

Vittoria Lanzo and Dico Angelo. Sentence-Length Dispersion: Human Articles and Model Generations. Hugging Face dataset, revision 40c03dde536decc008740b000a020d49782a567a, 2026. URL https://huggingface.co/datasets/burstiness-engine/sentence-length-dispersion.

Aman Madaan, Niket Tandon, Prakhar Gupta, et al. Self-Refine: Iterative Refinement with Self-Feedback. NeurIPS 2023, 2023. URL https://arxiv.org/abs/2303.17651.

Jiří Milička, Anna Marklová, and Václav Cvrček. Benchmark of Stylistic Variation in LLM-Generated Texts. arXiv, 2025. URL https://arxiv.org/abs/2509.10179.

Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. Rethinking the Role of Demonstrations: What Makes In-Context Learning Work? EMNLP 2022, 2022. URL https://arxiv.org/abs/2202.12837.

Tara Radvand, Mojtaba Abdolmaleki, Mohamed Mostagir, and Ambuj Tewari. A Training-free Method for LLM Text Attribution. arXiv, 2025. URL https://arxiv.org/abs/2501.02406.

Alexander Shypula, Shuo Li, Botong Zhang, Vishakh Padmakumar, Kayo Yin, and Osbert Bastani. Evaluating the Diversity and Quality of LLM Generated Content. arXiv, 2025. URL https://arxiv.org/abs/2504.12522.

İsmail Tarım and Aytuğ Onan. Can You Detect the Difference? arXiv, 2025. URL https://arxiv.org/abs/2507.10475.

Edward Tian. Perplexity, burstiness, and statistical AI detection. GPTZero blog, 1 March 2023, 2023. URL https://gptzero.me/news/perplexity-and-burstiness-what-is-it/. Accessed 18 October 2026.

Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. Plan-and-Solve Prompting: Improving Zero-Shot Chain-of-Thought Reasoning by Large Language Models. ACL 2023, 2023. URL https://arxiv.org/abs/2305.04091.

Wikinews. Wikinews: free-content news source (English, 2016-2021), 2021. URL https://en.wikinews.org.

C. B. Williams. A note on the statistical analysis of sentence-length as a criterion of literary style. *Biometrika*, 1940. URL https://academic.oup.com/biomet/article-pdf/31/3-4/356/499025/31-3-4-356.pdf.

G. Udny Yule. On sentence-length as a statistical characteristic of style in prose. *Biometrika*, 1939. URL https://pure.mpg.de/pubman/item/item_2407783_2/component/file_2407782/Yule_1939_On_Sentence_Lenght_Biometrika_1939.pdf.

A Prompt wordings

The five prompting modes share one length-matched task brief and, apart from the control mode, one persona frame. This appendix gives the exact wording of every component, the rule that composes them into each mode, and two rules the wording leaves implicit: how the target length is set and how the few-shot exemplars are chosen. The byte-exact source is `bench/prompts.py` in the released code, which also carries a self-check that enforces the three-call structure of the self-refine mode and the metric-free property stated below. In the blocks that follow, lines beginning with # are annotations; every other line is the literal prompt string, with the long single-line strings wrapped to fit the page and blank lines marking the paragraph breaks inside a string. Braced tokens such as `{headline}` are filled per generation.

```
# BRIEF: control mode sends only this
Write a news article with the following headline:

{headline}

Aim for about {target_words} words.

# VOICE: prepended to every non-control mode
Act as a professional news journalist. Provide the article a seasoned
staff reporter would write: concrete and specific, in a natural human
voice, never flat or mechanical.

# TAIL: appended to every non-control mode
Keep that natural, human quality all the way through, including the
closing paragraphs.

# PLAN: plan-mode body, between VOICE and BRIEF
Before you write, plan the article the way a reporter does: pick the
lede (the single most important fact), then lay out the who, what,
when, where, why, and how, and the order to present them, most
important first. Work that outline out for yourself, then write the
article to it. Do not show your planning; return only the finished
article.

# FEWSHOT_LEAD / FEWSHOT_TAIL: wrap the exemplar block
Study the voice, pacing, and rhythm of the examples below, then write
in the same spirit. Borrow only their style, never their facts or
wording.

<examples>
<example>
# exemplar 1 (full article text omitted; see "Few-shot exemplars" below)
</example>
# ... two more <example> blocks, three exemplars in total ...

</examples>
```

```
# FEEDBACK: self-refine call 2, over call 1's {draft}
You are a demanding editor. Critique the draft below only on how
naturally human its prose reads: its rhythm, pacing, and flow. Name
at least THREE specific passages that read as flat, monotonous, or
mechanical. For each, quote the exact sentence(s) and say concretely
what to change to give it a more human cadence. Do not rewrite the
draft; give only the critique. Even if the writing is already strong,
still name the three weakest spots.

DRAFT:
{draft}

# REWRITE: self-refine call 3, over {draft} and {feedback}
Here is your draft and an editor's critique.

DRAFT:
{draft}

CRITIQUE:
{feedback}

Rewrite the article in full, applying every point of the critique. This
is a substantial rewrite, not a light edit: change how the sentences
move wherever the critique flagged flatness, while preserving the
meaning, the facts, and the target length. Return only the rewritten
article.
```

How the modes compose. The control mode sends the bare brief and nothing else. Every other mode follows one template: the voice preamble, then a mode-specific body, then the same brief, then the tail. The plain instruction has an empty body, so it reduces to the voice preamble, the brief, and the tail. The plan mode puts the plan text in the body slot, and the few-shot mode puts the exemplar block there. Self-refine is three model calls on a single article: a draft from the instruction prompt, a critique from the feedback prompt applied to that draft, and a rewrite from the rewrite prompt applied to the draft and the critique. A self-check in `bench/prompts.py` asserts the three calls.

Length brief. The `{target_words}` slot is the paired human article's word count rounded to the nearest 25, so every generation targets its human counterpart's length (Section 4.1).

Few-shot exemplars. The exemplar block holds three straight news reports drawn from Wikinews articles outside the 200 evaluation set. They are the reports whose burstiness sits nearest the median of the human distribution, chosen for register and topic. The selection uses B_n only to locate that median and drop degenerate tails, never to pick high-dispersion exemplars, so the few-shot condition sees typical human sentence-length dispersion and no exemplar is chosen to teach to the test. Their full texts are committed with the released code and are omitted here for length.

Metric-free wording. No prompt names the statistic the benchmark scores or gives a numeric target on it: a self-check enforces that no prompt string contains a measured-axis term, among them *burstiness*, *sentence length*, *variance* and *standard deviation*, and the word count in the length brief is a length target only. Every non-control prompt asks for prose that is never flat or mechanical; the few-shot and self-refine prompts name rhythm, pacing, flow and cadence, and the rewrite step says to change how the sentences move, as a user asking for human-sounding prose would. What the wording withholds is the measured quantity and any target on it, not the intent.

Instruction paraphrases (wording-robustness probe). The probe in Section 6 re-runs the instruction mode on the 18 October DeepSeek V4 Pro snapshot (DeepSeek V4 Pro† in the tables) under three independent rewrites of the voice and tail above, one sample over all 200 briefs, each keeping the persona frame and naming no statistic. The harness is `experiments/ablation/prompt_`

paraphrase.py. The three pairs follow, the voice prepended and the tail appended around the same brief.

```
# PARAPHRASE v1 (VOICE then TAIL)
You are a veteran newspaper reporter. Write the story the way
an experienced newsroom professional would: specific, grounded
in facts, with a voice that reads as unmistakably human rather
than machine-written.

Hold that authentic reporter's voice through to the final sentence.

# PARAPHRASE v2 (VOICE then TAIL)
Take the role of a seasoned journalist at a major daily. Produce
the kind of article an editor would run with light touches only:
concrete details, a living, natural voice, nothing stiff or
formulaic.

Make sure the writing still sounds fully human in the closing
paragraphs, not just the opening.

# PARAPHRASE v3 (VOICE then TAIL)
Write as a long-time staff correspondent. The article should
carry the texture of real human reportage: precise facts,
natural phrasing, and none of the flat, mechanical quality of
generated text.

Sustain that human texture to the very end of the piece.
```

B Per-model result grids

Every model ran the identical full design: the same 200 matched headlines, the same five mode wordings (Appendix A), three samples per cell, uniform normalization, and the same paired scoring on all four dimensions. Qwen3.6-27B's grid is Table 2, the grids that follow complete the set, and DeepSeek V4 Pro† (Appendix D) has its own grid beside its July leg (Table 9); all are generated from each leg's released evaluation with no hand-transcribed values. Within each model we apply Benjamini–Hochberg at $q = 0.05$ across its twenty cells and report the census in the caption, with the stricter Holm count alongside, and no correction is applied across models. Instruction-mode equivalence for every model is in Table 12 (Appendix D); Kimi K2.6's instructed leg shows no per-cell significant difference on any dimension, though only its dispersion cell carries an equivalence bound. Memory M is where six few-shot legs overshoot the human articles even as their dispersion stays significantly flat. Seg. entropy, the axis most exposed to formatting artifacts, still separates on normalized text: four of DeepSeek V4 Pro's five modes and four of Grok 4.5's five modes sit significantly above the human articles.

Table 2: Qwen3.6-27B, full grid ($n = 200$ headlines). Seed-averaged Wilcoxon signed-rank paired tests over the matched headlines; median Δ = human − model, so positive means the model sits *below* the human article on that dimension. Matched rank-biserial in parentheses; $^{*}p < 0.05$, $^{**}p < 0.01$ (per cell, two-sided). Within-model census: 14 of 20 cells per-cell significant; 14 survive Benjamini–Hochberg at $q = 0.05$ and 12 survive Holm.

Mode	B_n	Gini	Seg. entropy	Memory M
Control	0.102 $(0.70)^{**}$	0.051 $(0.70)^{**}$	-0.001 (0.10)	-0.017 (-0.10)
Instruction	0.035 $(0.35)^{**}$	0.010 $(0.29)^{**}$	-0.018 (-0.04)	-0.057 $(-0.20)^{*}$
Few-shot	0.119 $(0.80)^{**}$	0.060 $(0.79)^{**}$	0.127 $(0.47)^{**}$	-0.071 $(-0.26)^{**}$
Plan	0.043 $(0.42)^{**}$	0.017 $(0.36)^{**}$	0.057 (0.16)	-0.063 $(-0.18)^{*}$
Self-refine	0.041 $(0.27)^{**}$	0.021 $(0.25)^{**}$	0.029 (0.10)	0.023 (0.05)

Table 2 reports Qwen3.6-27B's paired deltas. The uninstructed control differs from the human articles where dispersion is measured: median $\Delta B_n = 0.102$ (rank-biserial 0.70, $p < 10^{-5}$), with

Table 3: Gemini 3.1 Pro, full grid ($n = 200$ headlines). Design and conventions as in Table 2. Within-model census: 16 of 20 cells per-cell significant; 16 survive Benjamini–Hochberg at $q = 0.05$ and 13 survive Holm.

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.128\ (0.86)^{**}$	$0.056\ (0.82)^{**}$	$-0.062\ (-0.19)^{*}$	$0.041\ (0.17)^{*}$
Instruction	$0.071\ (0.56)^{**}$	$0.026\ (0.47)^{**}$	$-0.053\ (-0.16)$	$0.109\ (0.37)^{**}$
Few-shot	$0.169\ (0.94)^{**}$	$0.080\ (0.93)^{**}$	$0.117\ (0.39)^{**}$	$0.051\ (0.24)^{**}$
Plan	$0.071\ (0.58)^{**}$	$0.028\ (0.48)^{**}$	$-0.026\ (-0.06)$	$0.112\ (0.46)^{**}$
Self-refine	$0.039\ (0.28)^{**}$	$0.013\ (0.21)^{**}$	$-0.051\ (-0.12)$	$0.046\ (0.12)$

Table 4: GLM-5.2, full grid ($n = 198/198/197/197/200$ headlines per mode, in Table 1 order). Design and conventions as in Table 2. Within-model census: 16 of 20 cells per-cell significant; 16 survive Benjamini–Hochberg at $q = 0.05$ and 14 survive Holm. Silent-refusal empty cells are excluded by the seed-averaging protocol (Section 4.4).

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.126\ (0.85)^{**}$	$0.058\ (0.83)^{**}$	$0.068\ (0.25)^{**}$	$0.053\ (0.19)^{*}$
Instruction	$0.064\ (0.58)^{**}$	$0.030\ (0.51)^{**}$	$0.064\ (0.26)^{**}$	$0.078\ (0.24)^{**}$
Few-shot	$0.148\ (0.94)^{**}$	$0.068\ (0.92)^{**}$	$0.221\ (0.74)^{**}$	$0.022\ (0.07)$
Plan	$0.081\ (0.66)^{**}$	$0.037\ (0.59)^{**}$	$0.091\ (0.32)^{**}$	$0.030\ (0.20)^{*}$
Self-refine	$0.012\ (0.15)$	$0.005\ (0.10)$	$0.050\ (0.33)^{**}$	$-0.023\ (-0.03)$

Table 5: Claude Opus 4.8, full grid ($n = 200$ headlines). Design and conventions as in Table 2. Within-model census: 16 of 20 cells per-cell significant; 16 survive Benjamini–Hochberg at $q = 0.05$ and 15 survive Holm. Ran at effort high with a 32,000-token cap (Section 4.4).

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.119\ (0.80)^{**}$	$0.054\ (0.77)^{**}$	$0.116\ (0.39)^{**}$	$-0.069\ (-0.24)^{**}$
Instruction	$-0.037\ (-0.25)^{**}$	$-0.031\ (-0.36)^{**}$	$0.032\ (0.09)$	$-0.117\ (-0.38)^{**}$
Few-shot	$0.039\ (0.34)^{**}$	$0.012\ (0.26)^{**}$	$0.253\ (0.75)^{**}$	$-0.045\ (-0.25)^{**}$
Plan	$0.000\ (0.01)$	$-0.004\ (-0.10)$	$0.108\ (0.34)^{**}$	$-0.059\ (-0.25)^{**}$
Self-refine	$-0.065\ (-0.50)^{**}$	$-0.043\ (-0.59)^{**}$	$-0.005\ (0.01)$	$0.039\ (0.18)^{*}$

Table 6: Kimi K2.6, full grid ($n = 199/199/199/199/198$ headlines per mode, in Table 1 order). Design and conventions as in Table 2. Within-model census: 12 of 20 cells per-cell significant; 11 survive Benjamini–Hochberg at $q = 0.05$ and 11 survive Holm (lost under Benjamini–Hochberg: control Seg. entropy). Provider content-filter refusals are excluded by the seed-averaging protocol (Section 4.4).

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.106\ (0.73)^{**}$	$0.050\ (0.69)^{**}$	$-0.048\ (-0.16)^{*}$	$-0.091\ (-0.28)^{**}$
Instruction	$-0.007\ (0.00)$	$-0.010\ (-0.14)$	$-0.024\ (-0.11)$	$-0.026\ (-0.03)$
Few-shot	$0.133\ (0.85)^{**}$	$0.064\ (0.81)^{**}$	$0.152\ (0.55)^{**}$	$-0.083\ (-0.28)^{**}$
Plan	$0.029\ (0.25)^{**}$	$0.008\ (0.10)$	$0.032\ (0.05)$	$0.045\ (0.09)$
Self-refine	$-0.069\ (-0.52)^{**}$	$-0.041\ (-0.54)^{**}$	$0.061\ (0.16)$	$-0.064\ (-0.27)^{**}$

Table 7: Gemma 3 27B, full grid ($n = 200$ headlines). Design and conventions as in Table 2. Within-model census: 19 of 20 cells per-cell significant; 19 survive Benjamini–Hochberg at $q = 0.05$ and 19 survive Holm.

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.091\ (0.76)^{**}$	$0.048\ (0.75)^{**}$	$0.063\ (0.25)^{**}$	$-0.104\ (-0.29)^{**}$
Instruction	$0.057\ (0.47)^{**}$	$0.027\ (0.44)^{**}$	$0.086\ (0.36)^{**}$	$-0.088\ (-0.31)^{**}$
Few-shot	$0.081\ (0.68)^{**}$	$0.042\ (0.67)^{**}$	$0.145\ (0.48)^{**}$	$-0.124\ (-0.42)^{**}$
Plan	$0.070\ (0.62)^{**}$	$0.036\ (0.61)^{**}$	$0.129\ (0.45)^{**}$	$-0.076\ (-0.30)^{**}$
Self-refine	$0.045\ (0.41)^{**}$	$0.017\ (0.39)^{**}$	$0.094\ (0.32)^{**}$	$-0.014\ (-0.04)$

Table 8: DeepSeek V4 Pro, full grid ($n = 200$ headlines). Design and conventions as in Table 2. Within-model census: 12 of 20 cells per-cell significant; 12 survive Benjamini–Hochberg at $q = 0.05$ and 11 survive Holm. The leg's cap was raised from 16,000 to 32,000 tokens early in the run (Appendix D.3).

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.050\ (0.48)^{**}$	$0.029\ (0.47)^{**}$	$-0.050\ (-0.12)$	$-0.084\ (-0.29)^{**}$
Instruction	$0.003\ (0.06)$	$-0.004\ (0.00)$	$-0.145\ (-0.47)^{**}$	$-0.057\ (-0.25)^{**}$
Few-shot	$0.032\ (0.36)^{**}$	$0.015\ (0.33)^{**}$	$-0.068\ (-0.26)^{**}$	$-0.042\ (-0.26)^{**}$
Plan	$0.012\ (0.16)$	$0.001\ (0.09)$	$-0.117\ (-0.36)^{**}$	$-0.064\ (-0.22)^{**}$
Self-refine	$-0.002\ (0.04)$	$-0.001\ (0.02)$	$-0.139\ (-0.49)^{**}$	$-0.007\ (-0.06)$

Table 9: DeepSeek V4 Pro†, full grid ($n = 200$ headlines). Design and conventions as in Table 2. Within-model census: 11 of 20 cells per-cell significant; 9 survive Benjamini–Hochberg at $q = 0.05$ and 9 survive Holm (lost under Benjamini–Hochberg: plan B_n, instruction Seg. entropy). The DeepSeek V4 Pro leg re-run on 18 October 2026 against the snapshot the API alias then served, with the same prompts and briefs (Appendix D); reported beside the nine models and outside their counts.

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.039\ (0.34)^{**}$	$0.018\ (0.29)^{**}$	$0.023\ (0.14)$	$-0.070\ (-0.24)^{**}$
Instruction	$-0.037\ (-0.27)^{**}$	$-0.027\ (-0.34)^{**}$	$-0.060\ (-0.18)^{*}$	$-0.013\ (-0.10)$
Few-shot	$0.018\ (0.14)$	$0.004\ (0.04)$	$0.132\ (0.29)^{**}$	$-0.003\ (0.03)$
Plan	$-0.030\ (-0.18)^{*}$	$-0.022\ (-0.25)^{**}$	$-0.018\ (0.01)$	$-0.036\ (-0.09)$
Self-refine	$-0.042\ (-0.29)^{**}$	$-0.020\ (-0.30)^{**}$	$0.001\ (-0.05)$	$0.011\ (0.05)$

Table 10: Grok 4.3, full grid ($n = 200$ headlines). Design and conventions as in Table 2. Within-model census: 16 of 20 cells per-cell significant; 16 survive Benjamini–Hochberg at $q = 0.05$ and 13 survive Holm.

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.052\ (0.45)^{**}$	$0.024\ (0.43)^{**}$	$-0.046\ (-0.16)$	$-0.090\ (-0.28)^{**}$
Instruction	$0.065\ (0.57)^{**}$	$0.033\ (0.53)^{**}$	$0.062\ (0.19)^{*}$	$-0.039\ (-0.20)^{*}$
Few-shot	$0.116\ (0.83)^{**}$	$0.055\ (0.80)^{**}$	$0.180\ (0.56)^{**}$	$-0.019\ (-0.13)$
Plan	$0.082\ (0.65)^{**}$	$0.038\ (0.60)^{**}$	$0.136\ (0.35)^{**}$	$-0.026\ (-0.09)$
Self-refine	$0.039\ (0.35)^{**}$	$0.017\ (0.30)^{**}$	$0.055\ (0.21)^{**}$	$-0.008\ (0.02)$

Table 11: Grok 4.5, full grid ($n = 200$ headlines). Design and conventions as in Table 2. Within-model census: 12 of 20 cells per-cell significant; 12 survive Benjamini–Hochberg at $q = 0.05$ and 8 survive Holm.

Mode	B_n	Gini	Seg. entropy	Memory M
Control	$0.002\ (-0.01)$	$0.003\ (-0.05)$	$-0.249\ (-0.74)^{**}$	$-0.079\ (-0.22)^{**}$
Instruction	$-0.022\ (-0.13)$	$-0.015\ (-0.14)$	$-0.099\ (-0.40)^{**}$	$-0.017\ (-0.12)$
Few-shot	$0.106\ (0.77)^{**}$	$0.050\ (0.76)^{**}$	$0.089\ (0.28)^{**}$	$-0.047\ (-0.20)^{*}$
Plan	$0.004\ (0.05)$	$0.000\ (0.03)$	$-0.062\ (-0.22)^{**}$	$-0.058\ (-0.23)^{**}$
Self-refine	$-0.034\ (-0.30)^{**}$	$-0.026\ (-0.34)^{**}$	$-0.133\ (-0.44)^{**}$	$-0.001\ (0.06)$

Gini concurring (0.051, rank-biserial 0.70). On segmentation entropy (-0.001, $p = 0.24$) and memory (-0.017, $p = 0.20$) the control mode shows no significant difference. The control-mode authorship gap is significant only on the dispersion dimensions; the memory and entropy intervals ($[-0.073, 0.017]$ and $[-0.050, 0.063]$) each clear only the 0.08 margin.

A plain instruction raises dispersion significantly relative to the bare brief (paired contrast 0.049, $p < 10^{-5}$) and does not close the gap: ΔB_n falls from 0.102 to 0.035 (rank-biserial 0.35, $p = 2 \times 10^{-5}$) and Gini from 0.051 to 0.010 ($p = 3 \times 10^{-4}$); both effects survive correction. Against the equivalence margins, instruction-mode B_n is equivalent at the 0.08 margin (CI $[0.010, 0.052]$, $m^* = 0.052$); Gini is equivalent at the 0.03 margin ($m^* = 0.025$), segmentation entropy equivalent at the 0.08 margin ($m^* = 0.064$, CI $[-0.064, 0.046]$), and memory equivalent at none of the pre-specified margins ($m^* = 0.119$, CI $[-0.119, -0.016]$); the same numeric margins are applied to the companions, whose scales differ, and the calibration of Section 4.3 is for B_n only.

Few-shot exemplars are the flattest mode on the dispersion statistics by median and rank-biserial, a B_n deficit of 0.119 with rank-biserial 0.80 (significantly flatter than instruction by direct contrast, -0.077, $p < 10^{-5}$), and the per-seed effect holds at 0.72–0.74 on every individual seed.

Plan and self-refine behave like weaker instructions: significant dispersion deficits (B_n 0.043 and 0.041, rank-biserial 0.42 and 0.27, both $p < 0.01$) with no correction-surviving effect on segmentation entropy (plan 0.057, $p = 0.052$; self-refine 0.029, $p = 0.22$); on memory, plan's overshoot (-0.063, $p = 0.02$) survives correction and self-refine shows none (0.023, $p = 0.52$). All ten dispersion cells (five modes by B_n and Gini) are significant at $p < 0.01$ and survive correction; the other survivors are few-shot's segmentation entropy (0.127) and memory (-0.071) cells and the instruction and plan memory overshoots.

C Reference implementation

The measurement instrument is model-free and pure standard library. The listing below is the complete, self-contained implementation used in this paper: the sentence segmenter, the burstiness coefficient B_n (Goh–Barabási under the Kim–Jo finite-n correction), the memory coefficient M, the Gini robustness check, Miller–Madow segmentation entropy, the multiscale profile, and the Goodhart guard, whose thresholds are its default keyword arguments. It runs on any Python 3 interpreter with no third-party dependencies and no reference model. Three conventions the prose leaves to the listing: σ in $r = \sigma/\mu$ is the population standard deviation, the Gini divides by $n(n-1)$, and segmentation entropy is in bits over the marks `. ! ? , ; : - ( ) " '` and the em dash, after typographic quotes are mapped to ASCII.

```
"""Reference implementation of the burstiness measurement instrument.

Model-free and pure standard library. Segments a document into sentence
lengths and computes the burstiness coefficient B_n (Goh-Barabasi under the
Kim-Jo finite-n correction), the memory coefficient M, the Gini robustness
check, Miller-Madow segmentation entropy, and a multiscale profile. The
Goodhart guard screens degenerate text that would game a dispersion score.
"""
from __future__ import annotations

import math
import re
from statistics import fmean, pstdev
from typing import List, Optional, Sequence

# Below this many sentences, a single document's B_n and M estimates are
# noisy; the entry point reports ``reliable = False``.
RELIABILITY_FLOOR = 10

# ---------------------------------------------------------------------------
# Segmentation: sentence lengths in whitespace tokens (stdlib only)
# ---------------------------------------------------------------------------

_ABBREVIATIONS = {
    "mr", "mrs", "ms", "dr", "prof", "sr", "jr", "st", "vs", "etc",
    "e.g", "i.e", "fig", "no", "vol", "cf", "al", "inc", "ltd", "co",
```

```
}
_SENT_SPLIT = re.compile(r'([.!?]+["\')\]]?)\s+')
_PUNCT_SYMBOLS = re.compile(r'[.!?,;:\-\u2014()"\']')
_TYPO_QUOTES = str.maketrans({"\u201c": '"', "\u201d": '"', "\u2018": "'", "\u2019": "'"})


def sentence_lengths(text: str) -> List[int]:
    """Word-token count of each sentence, with an abbreviation-guarded split."""
    if not text or not text.strip():
        return []
    text = text.translate(_TYPO_QUOTES)
    sentences: List[str] = []
    last = 0
    for m in _SENT_SPLIT.finditer(text):
        end = m.end()
        candidate = text[last:end].strip()
        bare = candidate.rstrip('.!?"\')]').strip()
        last_word = bare.split()[-1].lower() if bare.split() else ""
        if last_word in _ABBREVIATIONS:
            continue
        if candidate:
            sentences.append(candidate)
        last = end
    tail = text[last:].strip()
    if tail:
        sentences.append(tail)
    return [len(s.split()) for s in sentences if s.split()]


# ---------------------------------------------------------------------------
# Core coefficients (operate on a sequence of sentence lengths)
# ---------------------------------------------------------------------------

def burstiness_B(intervals: Sequence[float]) -> Optional[float]:
    """Goh-Barabasi coefficient B = (sigma - mu) / (sigma + mu), bounded [-1, 1]."""
    if len(intervals) < 2:
        return None
    mu = fmean(intervals)
    sigma = pstdev(intervals)
    denom = sigma + mu
    if denom <= 0:
        return None
    return (sigma - mu) / denom


def burstiness_B_corrected(intervals: Sequence[float]) -> Optional[float]:
    """Kim-Jo finite-n correction B_n; reduces to (r-1)/(r+1) as n -> inf."""
    n = len(intervals)
    if n < 2:
        return None
    mu = fmean(intervals)
    if mu <= 0:
        return None
    r = pstdev(intervals) / mu
    sp, sm = math.sqrt(n + 1), math.sqrt(n - 1)
    denom = (sp - 2.0) * r + sm
    if denom == 0:
        return None
    return (sp * r - sm) / denom


def memory_M(intervals: Sequence[float]) -> Optional[float]:
    """Lag-1 autocorrelation of consecutive sentence lengths (the memory term)."""
    n = len(intervals)
    if n < 3:
        return None
    a = list(intervals[:-1])
    b = list(intervals[1:])
```

```
    m1, m2 = fmean(a), fmean(b)
    s1, s2 = pstdev(a), pstdev(b)
    if s1 <= 0 or s2 <= 0:
        return None
    cov = fmean((x - m1) * (y - m2) for x, y in zip(a, b))
    return cov / (s1 * s2)


def memory_M_corrected(intervals: Sequence[float]) -> Optional[float]:
    """memory_M plus 1/(n-1), n the number of intervals: the M the benchmark scores."""
    m = memory_M(intervals)
    return None if m is None else m + 1.0 / (len(intervals) - 1)


def gini(intervals: Sequence[float]) -> float:
    """Gini coefficient of sentence lengths; scale-free robustness check for B_n."""
    n = len(intervals)
    if n < 2:
        return 0.0
    m = fmean(intervals)
    if m == 0:
        return 0.0
    # Sample (unbiased) Gini: divide by n*(n-1), not n^2, which removes an
    # n-dependent bias that would confound cross-document comparison.
    return sum(abs(a - b) for a in intervals for b in intervals) / (2.0 * n * (n - 1) * m)


def segmentation_entropy(text: str) -> float:
    """Miller-Madow-corrected Shannon entropy (bits) of the punctuation marks.

    Typographic quotes are normalized to ASCII first, so quote STYLE (curly vs
    straight) is not scored as segmentation diversity.
    """
    counts: dict = {}
    for ch in _PUNCT_SYMBOLS.findall((text or "").translate(_TYPO_QUOTES)):
        counts[ch] = counts.get(ch, 0) + 1
    total = sum(counts.values())
    if total == 0:
        return 0.0
    h = -sum((c / total) * math.log2(c / total) for c in counts.values())
    return h + (len(counts) - 1) / (2.0 * total * math.log(2))


# ---------------------------------------------------------------------------
# Multiscale temporal profile
# ---------------------------------------------------------------------------

def _coarse_grain(L: Sequence[float], w: int) -> List[float]:
    """Non-overlapping window means: coarsen the length series to scale w."""
    return [fmean(L[i:i + w]) for i in range(0, len(L) - w + 1, w)]


def multiscale_burstiness(text: str, scales: Sequence[int] = (1, 2, 4)) -> dict:
    """B_n at the sentence scale and at coarse-grained window scales.

    Genuine long-range burstiness holds its dispersion at coarse scales, whereas
    mechanical alternation washes out toward B_n = -1 at scale >= 2.
    """
    L = [float(x) for x in sentence_lengths(text)]
    out = {}
    for s in scales:
        cg = list(L) if s == 1 else _coarse_grain(L, s)
        out[f"B_n@{s}"] = burstiness_B_corrected(cg) if len(cg) >= 2 else None
    return out


# ---------------------------------------------------------------------------
# Goodhart guard: flag degenerate texts that reach the score by gaming it
```

```
# ---------------------------------------------------------------------------

def _trigram_diversity(text: str) -> float:
    """Distinct word-trigram ratio (catches verbatim repetition)."""
    w = (text or "").split()
    if len(w) < 4:
        return 1.0
    tris = list(zip(w, w[1:], w[2:]))
    return len(set(tris)) / len(tris)


def _runon_ratio(text: str) -> float:
    """Internal clause delimiters (, ; :) per sentence terminal (. ! ?)."""
    internal = len(re.findall(r"[,;:]", text or ""))
    terminal = len(re.findall(r"[.!?]+", text or "")) or 1
    return internal / terminal


def gaming_flags(text, *, loo_delta=0.25, m_floor=-0.5, m_ceiling=0.7,
                 distinct_len_min=0.5, trigram_min=0.5, frag_frac_max=0.3,
                 punct_rate_max=0.25, runon_max=8.0, finescale_drop=0.4):
    """Flag texts that reach a high dispersion score through degenerate patterns.

    Returns {"ok", "gate_ok", "flags", "detail"}. ``ok`` means no gaming pattern
    fired; ``gate_ok`` means ok AND reliable (n >= RELIABILITY_FLOOR). A use that
    selects text by B_n should require gate_ok and a separate quality gate:
    re-punctuating a fixed text can swing B_n across much of its range, so B_n
    must never select generations on its own.

    Flags: outlier_driven (dropping the longest sentence swings B_n > loo_delta),
    low_len_diversity (distinct-length ratio < distinct_len_min at n >= 8, which
    catches blocky and period-k patterns), repetitive (trigram ratio <
    trigram_min), alternating (M < m_floor), over_clumped (M > m_ceiling),
    fine_scale_only (B_n@1 > 0 but drops by > finescale_drop at scale 2),
    fragment_spam (>= frag_frac_max of sentences under 3 words), punct_gamed
    (punctuation marks per word > punct_rate_max), runon (internal-to-terminal
    delimiter ratio > runon_max).
    """
    Li = [int(x) for x in sentence_lengths(text)]
    L = [float(x) for x in Li]
    n = len(L)
    if n < 3:
        return {"ok": False, "gate_ok": False, "flags": {"too_short": True}, "detail":
     {"n": n}}

    B_full = burstiness_B_corrected(L)
    j = max(range(n), key=lambda i: L[i])
    B_loo = burstiness_B_corrected(L[:j] + L[j + 1:])
    loo = abs(B_full - B_loo) if (B_full is not None and B_loo is not None) else 0.0
    M = memory_M(L)  # the guard's thresholds are set on the uncorrected coefficient
    distinct_ratio = len(set(Li)) / n
    trig = _trigram_diversity(text)
    frag_frac = sum(1 for x in Li if x < 3) / n
    words = sum(L) or 1.0
    punct_rate = len(_PUNCT_SYMBOLS.findall(text or "")) / words
    runon = _runon_ratio(text)
    ms = multiscale_burstiness(text)
    b1, b2 = ms.get("B_n@1"), ms.get("B_n@2")
    fine_scale_only = (b1 is not None and b2 is not None and b1 > 0.0
                       and (b1 - b2) > finescale_drop)

    flags = {
        "outlier_driven": loo > loo_delta,
        "low_len_diversity": n >= 8 and distinct_ratio < distinct_len_min,
        "repetitive": trig < trigram_min,
        "alternating": M is not None and M < m_floor,
        "over_clumped": M is not None and M > m_ceiling,
        "fine_scale_only": fine_scale_only,
```

```
        "fragment_spam": frag_frac >= frag_frac_max,
        "punct_gamed": punct_rate > punct_rate_max,
        "runon": runon > runon_max,
    }
    ok = not any(flags.values())
    return {
        "ok": ok,
        "gate_ok": ok and n >= RELIABILITY_FLOOR,
        "flags": flags,
        "detail": {"n": n, "loo_dB": round(loo, 3),
                   "M": None if M is None else round(M, 3),
                   "distinct_ratio": round(distinct_ratio, 3), "trigram_div": round(trig,
 3),
                   "frag_frac": round(frag_frac, 3), "punct_rate": round(punct_rate, 3),
                   "runon": round(runon, 2)},
    }


# ---------------------------------------------------------------------------
# Entry point: the full model-free vector for a document
# ---------------------------------------------------------------------------

def burstiness_pp(text: str) -> dict:
    """Model-free structural burstiness of a document.

    B_n is the primary dispersion coefficient, M the lag-1 memory, Gini the
    robustness check, and multiscale the temporal profile. Segmentation entropy
    is a separate axis (see segmentation_entropy) and surprisal is kept outside
    the construct. ``reliable`` is False below RELIABILITY_FLOOR sentences.
    """
    L = [float(x) for x in sentence_lengths(text)]
    n = len(L)
    return {
        "B_n": burstiness_B_corrected(L),
        "M": memory_M_corrected(L),
        "gini": gini(L),
        "multiscale": multiscale_burstiness(text),
        "n_sentences": n,
        "reliable": n >= RELIABILITY_FLOOR,
        "sentence_lengths": [int(x) for x in L],
    }
```

D Protocol and audit detail

D.1 Corpus parse mechanics

The human side of each pair is the plain article body, regenerable from the source Wikinews wikitext by a deterministic MediaWiki parse (`experiments/ablation/clean_wikinews.py`) that keeps the body up to the first section heading or reader-comment block, removes templates, drops reference tags with their contents, resolves wikilinks to their display text, and unwraps inline formatting. On a held-out check of 100 articles this parse reproduces the shipped human bodies exactly.

D.2 Per-leg completeness

Completeness is disclosed per leg. Of Qwen3.6-27B's 3,000 cells (200 headlines $\times$ 5 modes $\times$ 3 seeds), 2,996 completed; we excluded the four that exceeded the reasoning budget rather than scoring their incomplete output. We average affected headlines over their remaining seeds, so every mode retains 200 matched pairs. Grok 4.5 reports two empty self-refine outputs, truncations by signature, seed-averaged away. Provider content moderation removes cells on two legs, by two different mechanisms. Moonshot's filter refuses explicitly: 25 of Kimi K2.6's 3,000 cells return HTTP 400, one Hong Kong protest headline refused at every mode and seed. Zhipu's filter refuses silently: 36 of GLM-5.2's cells return an ordinary HTTP 200 with empty output, concentrated on the same cluster of politically sensitive headlines, leaving 197–200 paired headlines per mode on the GLM-5.2 leg and

198–199 on the Kimi K2.6 leg (per-mode counts in the grid captions of Appendix B). The remaining five legs generated all 200 headlines in every mode.

D.3 Per-leg decoding and output caps

GLM-5.2 ignores a requested temperature and Kimi K2.6 rejects one; Grok 4.5 exposes neither temperature nor top-p; Gemini 3.1 Pro ran at temperature 1.0 with provider-managed thinking; Claude Opus 4.8 exposes no temperature. Gemini's thoughts are never returned, and Gemma 3 has no thinking mode. Output caps: $16{,}000$ tokens on the API legs, except DeepSeek V4 Pro, whose cap was raised from $16{,}000$ to $32{,}000$ after 51 of its first 513 cells spent the whole budget on reasoning (those cells were regenerated at the larger cap and the other early cells kept), and Claude Opus 4.8 at $32{,}000$; on the open-weight legs the cap is per article, $6{,}144$ tokens on Qwen3.6-27B, whose thinking shares the budget, and $1{,}024$ on Gemma 3 27B. Anthropic bills reasoning against the cap, so at $16{,}000$ tokens and maximum effort the Opus self-refine rewrite spent the whole budget reasoning and emitted no prose; the leg ran at high effort with the $32{,}000$-token cap, which restores a comparable prose budget rather than granting a larger one; Gemini 3.1 Pro ran at thinking level high. The Opus leg completed all $3{,}000$ cells with no truncated or empty output. Each cell was sampled three times: at seeds 0, 1, 2 on the open-weight legs, as independent calls on the API legs.

D.4 DeepSeek V4 Pro at a later snapshot

DeepSeek V4 Pro† re-ran the DeepSeek V4 Pro grid on 18 October 2026 through the API alias the July leg used (7 July). The alias was documented at the time as DeepSeek-V4-Pro-0813, a version dated 13 August, after the July run; the July rows record only the bare alias, so the version they were served is not named. Settings: the same 200 briefs and five prompts (every first-stage request matches `bench/prompts.py` on the frozen briefs), three samples per cell, thinking at reasoning effort maximum as in July, a $32{,}000$-token cap throughout (the July leg kept most of its early cells generated at $16{,}000$, Appendix D.3), no temperature, top-p or seed sent. All $3{,}000$ cells completed with a stop finish reason, some after retries of transport errors or empty results, documented in the released repair and continuation manifests [Lanzo and Angelo, 2026]. One self-refine rewrite (scenario 93, sample 1) answered off topic in one 51-word sentence; it is kept unaltered in the release and, falling below the scorer's five-sentence floor, is dropped from that headline's self-refine seed mean (the other two samples carry the headline). The paraphrase probe of Section 6 ran on the same day through the same alias at the same settings, two of its 600 cells after one retry; the version of this paper reviewed on OpenReview reported a July probe of the same design, re-run because its raw generations could not be recovered for re-scoring with the corrected segmenter. Paired on the same headlines, the October leg is more dispersed than the July leg in every mode: B_n deltas (October $-$ July) of 0.015 to 0.038, five of five significant, the largest $p = 9 \times 10^{-4}$. Its instruction, plan and self-refine cells land significantly past the human articles at the per-cell level (Table 1; the plan cell does not survive the within-leg correction, Table 9), where the July cells did not; its few-shot cell, significantly flatter than the human articles in July (0.032, $p = 1 \times 10^{-5}$), is not significantly different in October (0.018, $p = 0.09$), though still significantly flatter than the leg's instruction cell (Table 13). The leg is reported in every table of this appendix and in Figure 1; the censuses and counts of Section 4.4 are over the nine models.

D.5 Confidence intervals

CIs on median paired deltas are percentile bootstrap intervals: the 2.5th and 97.5th percentiles of the medians of $2{,}000$ resamples of the matched pairs. The matched rank-biserial is $(W^+ - W^-)/(W^+ + W^-)$ over the non-zero paired deltas, and a sample on which a metric is undefined is dropped from that headline's seed mean. Per-seed B_n effect sizes are released alongside the evaluation artifacts and are consistent with the seed-averaged results.

D.6 Scaffolding rates

A generation-hygiene guard flagged markdown scaffolding and echoed headline blocks in every model we benchmark, from 18% (markdown) and 40% (title lines) of control-mode generations on one leg (DeepSeek V4 Pro) to 100% on another (Gemini 3.1 Pro, both features). All of it is formatting the plain-prose human baseline lacks.

D.7 Matching and lengths

Within a pair the model sees the headline and the human article's word count rounded to the nearest 25; it does not see the article. Achieved length tracks the target: the median ratio of generated to human word count ranges from 0.92 to 1.18 across legs and modes (Table 15, which also gives each leg's median sentence count across modes). The human articles have 10 to 32 sentences (median 15); the scorer requires five, and in any one mode of a leg at most 23% of the scored generations have fewer than ten.

D.8 Equivalence for every model

Table 12 applies the pre-specified margins to every model's instruction-mode B_n cell with the two verdicts kept apart: the Wilcoxon test against zero, and the tightest margin the 95% CI clears (m^*). Margins are compared with the unrounded interval; printed bounds and m^* are rounded outward to three decimals, so a printed m^* at or below a margin means the interval clears it. DeepSeek V4 Pro's unrounded upper bound, 0.0302, lies within the bootstrap's resolution of the 0.03 margin; by this rule it does not clear 0.03, and the text counts it at 0.05. Gemma 3 27B's, 0.0802, likewise sits on the 0.08 margin and is counted as clearing none. The human B_n distribution has median -0.414, standard deviation 0.127 and interquartile range $[-0.506, -0.334]$ (inverted-CDF quantiles).

Table 12: Instruction mode, B_n: median paired delta (human − model), its 95% bootstrap CI, the Wilcoxon test against zero, m^*, and whether the CI sits inside each pre-specified margin.

Model	median ΔB_n	95% CI	Wilcoxon	m^*	0.03	0.05	0.08
Gemini 3.1 Pro	0.071	$[0.050, 0.090]$	$p < 10^{-5}$	0.090	–	–	–
GLM-5.2	0.064	$[0.047, 0.087]$	$p < 10^{-5}$	0.087	–	–	–
Claude Opus 4.8	-0.037	$[-0.061, -0.013]$	$p = 0.002$	0.061	–	–	✓
Qwen3.6-27B	0.035	$[0.010, 0.052]$	$p = 2 \times 10^{-5}$	0.052	–	–	✓
Kimi K2.6	-0.007	$[-0.033, 0.017]$	$p = 0.98$	0.033	–	✓	✓
Gemma 3 27B	0.057	$[0.037, 0.081]$	$p < 10^{-5}$	0.081	–	–	–
DeepSeek V4 Pro	0.003	$[-0.023, 0.031]$	$p = 0.47$	0.031	–	✓	✓
DeepSeek V4 Pro†	-0.037	$[-0.062, -0.021]$	$p = 8 \times 10^{-4}$	0.062	–	–	✓
Grok 4.3	0.065	$[0.038, 0.098]$	$p < 10^{-5}$	0.098	–	–	–
Grok 4.5	-0.022	$[-0.039, 0.006]$	$p = 0.10$	0.039	–	✓	✓

D.9 Direct mode contrasts

Table 13 tests the mode-to-mode statements of Section 4.4 directly: for each model, the seed-averaged B_n of one mode minus another on the same headlines, one Wilcoxon signed-rank test per contrast. A negative delta means the first mode writes flatter text than the second.

Table 13: Paired mode contrasts on B_n, median delta with matched rank-biserial in parentheses; $^*p < 0.05$, $^{**}p < 0.01$. Negative: the first mode is flatter.

Model	Instr. − control	Few-shot − control	Few-shot − instr.
Gemini 3.1 Pro	$0.055\ (0.73)^{**}$	$-0.047\ (-0.57)^{**}$	$-0.099\ (-0.93)^{**}$
GLM-5.2	$0.066\ (0.73)^{**}$	$-0.027\ (-0.38)^{**}$	$-0.078\ (-0.87)^{**}$
Claude Opus 4.8	$0.144\ (0.99)^{**}$	$0.070\ (0.79)^{**}$	$-0.067\ (-0.76)^{**}$
Qwen3.6-27B	$0.049\ (0.65)^{**}$	$-0.024\ (-0.36)^{**}$	$-0.077\ (-0.85)^{**}$
Kimi K2.6	$0.101\ (0.90)^{**}$	$-0.024\ (-0.40)^{**}$	$-0.148\ (-0.95)^{**}$
Gemma 3 27B	$0.047\ (0.66)^{**}$	$0.018\ (0.26)^{**}$	$-0.029\ (-0.53)^{**}$
DeepSeek V4 Pro	$0.057\ (0.69)^{**}$	$0.019\ (0.23)^{**}$	$-0.032\ (-0.52)^{**}$
DeepSeek V4 Pro†	$0.065\ (0.83)^{**}$	$0.028\ (0.40)^{**}$	$-0.048\ (-0.69)^{**}$
Grok 4.3	$-0.019\ (-0.26)^{**}$	$-0.068\ (-0.75)^{**}$	$-0.044\ (-0.63)^{**}$
Grok 4.5	$0.015\ (0.19)^{*}$	$-0.102\ (-0.95)^{**}$	$-0.117\ (-0.97)^{**}$

D.10 Raw variance beside B_n

Raw sentence-length variance is the obvious statistic for the sentence-length reading. On the 200 human articles, $\mathrm{var}(L)$ correlates 0.55 with mean sentence length and B_n correlates 0.00. Table 14 scores the same paired design with $\mathrm{var}(L)$: its verdicts follow length as well as dispersion and contradict B_n on several legs.

Table 14: Matched rank-biserial (human − model) for B_n and for raw $\mathrm{var}(L)$ under control and instruction; $^{*}p < 0.05$, $^{**}p < 0.01$. Positive: the model is below the human articles on that statistic.

	B_n		raw $\mathrm{var}(L)$	
Model	Control	Instruction	Control	Instruction
Gemini 3.1 Pro	0.86^{**}	0.56^{**}	0.46^{**}	0.10
GLM-5.2	0.85^{**}	0.58^{**}	0.60^{**}	0.30^{**}
Claude Opus 4.8	0.80^{**}	-0.25^{**}	0.49^{**}	-0.38^{**}
Qwen3.6-27B	0.70^{**}	0.35^{**}	0.80^{**}	0.47^{**}
Kimi K2.6	0.73^{**}	0.00	0.12	-0.52^{**}
Gemma 3 27B	0.76^{**}	0.47^{**}	0.49^{**}	0.14
DeepSeek V4 Pro	0.48^{**}	0.06	0.37^{**}	−0.11
DeepSeek V4 Pro†	0.34^{**}	-0.27^{**}	0.22^{**}	-0.22^{**}
Grok 4.3	0.45^{**}	0.57^{**}	0.45^{**}	0.34^{**}
Grok 4.5	−0.01	−0.13	0.05	0.00

D.11 Segmentation audit

The sentence segmenter splits on terminal punctuation followed by a space, optionally with an ASCII closing quote or bracket in between. The segmenter as first released did not list typographic closing quotes, so a sentence ending in ." followed by a space was not split from the next one. The human articles carry straight quotes and no such boundaries; several API models emit typographic quotes, with a leg mean of up to 1.3 missed boundaries per generation on the DeepSeek V4 Pro, Gemma 3 27B and Grok 4.5 legs, merges that on net make the model look more dispersed. The current version maps typographic quotes to ASCII before segmentation, and every benchmark number in this paper is scored with the corrected segmenter; the correction moved median ΔB_n by -0.002 to 0.029 and changed one of the forty-five landings (Table 15). Abbreviations outside the guard list, *U.S.* among them, cause a false split at rates of the same order on both sides (0.17 per human article, 0.07 to 0.26 per generation) and are left as they are.

D.12 Scoring sensitivity: floor and guard

Table 15 re-scores the B_n benchmark with a ten-sentence floor in place of the scorer's five, and Table 16 re-scores it without the generations the Goodhart guard flags and without the 8 human articles it flags (4%: one outlier-long sentence, heavy punctuation, strict alternation). The guard fires on 2–14% of a leg's generations in a given mode. Neither re-scoring is the main analysis: the five-sentence floor is the benchmark's choice and the ten-sentence flag the reference implementation's conservative default, and the guard was built to screen text selected by the score, which the benchmark does not do. The landing the guard filter changes is Grok 4.5's instruction cell (-0.022, $p = 0.10$ in the main scoring; -0.024, $p = 0.02$ without the flagged text), the cell the segmenter correction moved from significantly more dispersed to no significant difference; the ten-sentence floor changes Kimi K2.6's plan cell (0.029, $p = 0.002$, to 0.021, $p = 0.26$).

D.13 Contamination probe values

An n-gram contamination probe over 600 control-mode generations per model on three models finds mean matched 8-gram overlap with the paired article of 0.00055 (Qwen3.6-27B, max 0.036), 0.00207 (DeepSeek V4 Pro, max 0.104), and 0.00098 (Grok 4.3, max 0.062), against a mismatched-article baseline mean of 0.0 everywhere; 13-gram means are at or below 0.00064 for all three. The worst single generation shares roughly 10% of its 8-grams with its paired article, consistent with a shared quotation rather than memorization. Verbatim reproduction therefore cannot explain the measured

Table 15: Per model: B_n cells whose landing (flatter, no difference, more dispersed) changes under the segmenter correction and under a ten-sentence floor; the guard's flag rate across modes; the median, over individual generations, of the generated-to-human word-count ratio; the median sentence count across modes; and the largest share of scored generations under ten sentences in any mode. DeepSeek V4 Pro† postdates the segmenter correction and was scored only with the corrected segmenter (–).

	B_n cells changing landing					
Model	segmenter fix	10-sentence floor	guard flags	word ratio	sentences	<10 sentences
Gemini 3.1 Pro	0	0	6–14%	0.97–1.00	12–16	20%
GLM-5.2	0	0	6–8%	0.95–0.96	12–16	20%
Claude Opus 4.8	0	0	5–6%	0.99–1.05	13–17	11%
Qwen3.6-27B	0	0	4–8%	0.93–0.97	14–17	8%
Kimi K2.6	0	1	4–10%	1.04–1.11	11–18	23%
Gemma 3 27B	0	0	2–6%	1.05–1.14	14–17	6%
DeepSeek V4 Pro	0	0	2–6%	1.03–1.11	15–18	6%
DeepSeek V4 Pro†	–	0	6–9%	0.97–1.02	13–18	13%
Grok 4.3	0	0	4–6%	0.92–0.97	13–15	10%
Grok 4.5	1	0	2–5%	1.05–1.18	14–19	6%

Table 16: Main scoring / guard-filtered scoring: median ΔB_n per model and mode; $^{*}p < 0.05$, $^{**}p < 0.01$.

Model	Control	Instruction	Few-shot	Plan	Self-refine
Gemini 3.1 Pro	0.128^{**} / 0.118^{**}	0.071^{**} / 0.069^{**}	0.169^{**} / 0.169^{**}	0.071^{**} / 0.070^{**}	0.039^{**} / 0.039^{**}
GLM-5.2	0.126^{**} / 0.123^{**}	0.064^{**} / 0.059^{**}	0.148^{**} / 0.145^{**}	0.081^{**} / 0.079^{**}	0.012 / 0.005
Claude Opus 4.8	0.119^{**} / 0.107^{**}	-0.037^{**} / -0.048^{**}	0.039^{**} / 0.035^{**}	0.000 / -0.002	-0.065^{**} / -0.074^{**}
Qwen3.6-27B	0.102^{**} / 0.097^{**}	0.035^{**} / 0.028^{**}	0.119^{**} / 0.113^{**}	0.043^{**} / 0.042^{**}	0.041^{**} / 0.039^{**}
Kimi K2.6	0.106^{**} / 0.101^{**}	-0.007 / -0.007	0.133^{**} / 0.135^{**}	0.029^{**} / 0.021^{*}	-0.069^{**} / -0.077^{**}
Gemma 3 27B	0.091^{**} / 0.082^{**}	0.057^{**} / 0.047^{**}	0.081^{**} / 0.078^{**}	0.070^{**} / 0.066^{**}	0.045^{**} / 0.042^{**}
DeepSeek V4 Pro	0.050^{**} / 0.050^{**}	0.003 / 0.000	0.032^{**} / 0.029^{**}	0.012 / 0.009	-0.002 / -0.005
DeepSeek V4 Pro†	0.039^{**} / 0.032^{**}	-0.037^{**} / -0.040^{**}	0.018 / 0.014	-0.030^{*} / -0.032^{**}	-0.042^{**} / -0.044^{**}
Grok 4.3	0.052^{**} / 0.044^{**}	0.065^{**} / 0.055^{**}	0.116^{**} / 0.111^{**}	0.082^{**} / 0.074^{**}	0.039^{**} / 0.035^{**}
Grok 4.5	0.002 / 0.000	-0.022 / -0.024^{*}	0.106^{**} / 0.103^{**}	0.004 / 0.001	-0.034^{**} / -0.037^{**}

gaps on the three probed models, which span open-weights and API-served models. The probe covered three of the nine models, on the control mode only.